See Something, Say Something: Layered Driver Situational Awareness from Language

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***Abstract*—Drivers need to build and maintain sufficient Situational Awareness (SA) of their driving environment in order to drive safely. Therefore, the ability to estimate a driver's SA (or its lack thereof) is critical for effective driving assistance. Traditionally, a driver's SA has been modeled as the ability to recognize objects and spatial layouts. However, SA is layered, and encompasses not only perception and comprehension, but also *projection: what might happen next*. To this end, we create a novel dataset with 3,320 annotated layered SA labels from 42 participants observing a diverse set of driving videos in a supervisory control context. We extract layered SA labels directly from participants' natural language responses to SAGAT-style probes. Our analysis of how awareness varies due to high-level features about the gaze scanpaths, object properties, and distractions, uncovers systematic patterns in the depth and dynamics of SA. We also propose a simple and intuitive baseline to predict an observer's SA from gaze scanpaths, providing a first step towards computational modeling of layered SA. We will release the code and dataset publicly at https://layeredsituationalawareness.github.io/**

I. INTRODUCTION

A driver's situational awareness (SA) is their ability to perceive, comprehend, and predict the immediate future of their driving environment [1]. Maintaining adequate SA is crucial for safe navigation in complex traffic scenarios. Failure can lead to hazardous outcomes, such as missing a crossing pedestrian [2] or failing to notice a braking lead vehicle [3]. While advanced driver assistance systems (ADAS) capable of tracking 'important' objects [4] can supplement a driver's lack of situational awareness (SA), a richer understanding of the factors contributing to insufficient SA is still necessary. This understanding helps maximize the effectiveness of these systems and minimize intrusions, such as excessive or redundant alerts when the driver is already aware of a risk. Additionally, accurate SA estimation also enables ADAS to prepare drivers in a timely manner for safe control handovers in Level 3 autonomous vehicles [5]. Driver SA estimation is therefore crucial to enhance safety in shared-control for driving [6].

Prior work on modeling driver's SA has focused primarily on understanding the driver's perception of objects in the scene [7]–[9]. While perception of the road scene, and in particular hazards, is a crucial component of SA, it is not by itself sufficient for safe decision-making. Per Endsley's definition [1], SA extends beyond just a spatial understanding of individual objects and the environment, to include an understanding of interactions among agents and the environment, as well as a projection of their likely future states. We illustrate this "layered" definition in Fig. 1.

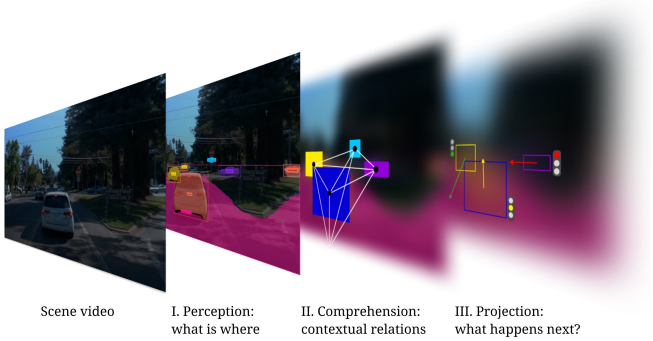


Fig. 1: **A conceptual illustration of layered Situational Awareness (SA)**. We follow [1] and posit that a driver's situational awareness of a scene progresses in stages, from (I) rapid perception of what is where, through (II) comprehension of context and how objects interact with one another, to (III) projection of possible future states through a complete understanding of the scene. In this work, we build the first dataset to study adequate and inadequate layered SA in normal driving, and investigate the performance of a simple model for layered SA prediction based on gaze and scene features.

In this work, we focus on modeling this more nuanced definition of SA. Specifically, we wish to study how eye gaze scanpaths on a driving scene inform a driver of a target object at the perception level (what is it), context level (where is it in relation to other objects and the environment), and future-projection level (what will it do next). To do so, we curate a novel dataset of real-world first-person driving scenarios, with eye gaze traces of drivers observing the scene, and ground-truth multi-layered SA labels for a target object.

The biggest challenge here lies in extracting ground-truth layered SA from drivers. Language serves as a natural interaction medium for humans to convey dense, high-level information. Furthermore, the advent of large language models (LLMs) has made downstream processing of large amounts of natural language data less labor-intensive. Thus, we collect layered SA labels as free-form natural language responses to SAGAT-style post-hoc probes [10] about a target object in the driving scenario. Fig. 2, shows two examples from our dataset of drivers scanning an identical driving scenario, highlighting

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the subtle differences in behavior at deeper layers of SA which may have safety consequences.

Additionally, to investigate the factors influencing layered SA, we define intuitive high-level features capturing (1) interactions between eye gaze and the target object, (2) the role of objects in the peripheral region, and (3) spatial and visual properties of the target object, to train simple classifier-based baselines as a first step towards computational modeling of layered situational awareness. To summarize our contributions:

- We propose a novel protocol for collecting ground-truth layered SA, using rich natural language feedback and LLMs to semi-automate annotation.
- We curate a novel dataset of gaze scanpaths over real-world driving scenarios, with $3,320$ ground truth layered SA annotations from 42 subjects.
- We analyze how a set of intuitive high-level eye gaze and scene-context features inform SA, and additionally provide simple baseline models for benchmarking.
- The code and dataset will be publicly released at: https://layeredsituationalawareness.github.io/

II. RELATED WORK

Situational awareness is a significant for maintaining safety in driving [11], [12]. Measuring, accommodating, and improving driver's SA are an important part of many intelligent vehicle systems, such as blind spot monitoring alerts and forward collision warnings [7], [13], and has been carefully explored in the context of information overload from assistive interfaces [14], [15] and control hand-backs during automated driving [16]–[19]. Research efforts into driver SA focus on several main challenges: (i) the accurate measurement of SA, in order to better understand its characteristics, and (ii) the creation of datasets from which to (iii) develop and evaluate models for SA prediction.

Measurement of Driver SA. SA describes a rich internal state of an agent and cannot be directly measured without significant intrusion. Attempts to measure it include passive behavioral techniques, such as through gaze fixations [20] and physiological signals [19], [21], or more intrusive probes, such as verbal or haptic queries to the driver [9], [22]. Probes have the advantage of allowing more direct access to the cognitive state of the driver, but are intermittent and cannot easily be deployed during real driving, while behavioral measurements are indirect but can be collected non-obtrusively. In our work, we use Situation Awareness Global Assessment Technique (SAGAT) [10] style probes to query participants observing a set of driving videos in a supervisory control context. SAGAT necessitates pausing the task at designated points (often in a simulated environment), to directly query the driver to recall particular aspects of the scene. While techniques such as Situation Present Assessment Method (SPAM) [23] have been used to probe drivers about scene details without interruption to the driving task, allowing them to be deployed during actual driving [24], these methods can lack temporal accuracy, since drivers can gain visual information after the probe has been initiated. Biswas et al. [9] proposed an interactive labeling protocol involving driver button presses to signal the detection of new objects in a driving scene. It captured continuous SA labels over time, but is challenging be extended to deeper layers of SA. Self-reporting techniques for SA such as SART [25] can be useful to gauge driver's experiences, but suffers from its own biases. Prior work using SAGAT in driving to measure SA [7] has only queries about the presence and location of objects.

Our work contrasts in a few key respects. Firstly, we query only specific target objects of interest, based on relevance to the ego vehicle and observable changes in their state (e.g. a pedestrian starting to cross a road, or a vehicle starting to change lanes). This limits the amount of information which drivers need to recall for a given query, reducing cognitive overload. Secondly, we allow subjects to respond in natural language, increasing the depth of their answers, and use LLMs to help interpret their responses at three different layers of SA (not just object presence, as in [7], [9], but also context and future prediction). Finally, we vary the observation times for a given stimulus across subjects, leading to negative as well as positive examples of SA.

Datasets for Driver SA. Developing models to predict driver SA that might be used in intelligent vehicle applications requires extensive, labeled datasets. To date, relatively few datasets have been made openly available to benchmark methods for driver SA modeling. Numerous public datasets which record driver gaze [26]–[28] or other behavioral data [29] during on-road driving exist, but do not include annotations for driver awareness. Third-party, post-hoc annotations of awareness can be applied to such data [8], but these annotations may be highly subjective and indirect. Public datasets which do measure first-hand awareness include the Personalized Situation Awareness of Drivers (PSAD) dataset [30], in which six experienced drivers observed real-world accident video footage and recorded their eye gaze and SAGAT responses at the point of responding to hazards in the scene, while Biswas et al. [9] used VR and a continuous-time button-press protocol to gather first-hand object awareness labels from twenty drivers as they navigated simulated traffic. In contrast, our dataset gathers gaze scanpaths over real-world driving footage from 42 subjects, with scenes chosen to reflect everyday (not necessarily hazardous) driving with varying levels of scene clutter [31], and with variations in probe timing to capture examples at different states of awareness. In addition, our natural language responses allow for richer labels covering all layers of situational awareness.

Models for Driver SA. Due to the scarcity of public driver SA datasets, there are relatively few approaches towards modeling driver SA from the behavioral observations of a driver, and cross-comparison of methods is difficult. Numerous efforts have shown the benefits and potential limitations of using eye gaze as the dominant cue for awareness [9], [20], [32], [33]. Advanced modeling frameworks incorporate human cognitive and behavioral aspects, such as human visual sensory and memory mechanisms [7], or using saliency-based methods [18]. Interpretable models for multi-class SA during condi-



Fig. 2: **Contrasting examples from the layered Situational Awareness (SA) dataset.** Here we illustrate two samples from our dataset, in which different subjects observed the same driving scene as though supervising an automated driving system. *Top*: the subject was aware of the black pick-up truck (shown in a red bounding box in the final frame) overtaking in the right of the scene, despite never fixating on it in preceding frames. They were therefore able to correctly judge the object's existence and context, but unable to describe its future behavior, when compared to the Oracle. *Bottom*: the subject did explicitly fixate on the pick-up truck, and hence was able to predict its behavior into the future. The layered SA dataset contains $3,320$ samples from 42 subjects and allows a more nuanced study into human visual information acquisition during driving supervision.

tionally automated driving have been explored [34], as well as multimodal approaches for SA measurement during takeover scenarios [19]. As our dataset explicitly provides SAGAT-labels at multiple layers of SA over a large number of subjects, we propose an existing feature-based SVM model [7] as a baseline for future experimentation.

III. DATA COLLECTION METHODOLOGY

We conducted a human subjects study to collect our layered SA dataset. Each subject was first shown a visual stimulus, their gaze was recorded, and were then asked to provide a verbal response indicating their existence, context, and future awareness of a target object from the stimulus. We employ a semi-automated data collection pipeline. Our final dataset has $3,320$ samples across 142 unique stimuli, awareness-responses, and eye gaze from 42 subjects. The following section describes each step of our procedure in detail and presents an analysis of the resulting dataset.

A. Stimulus Selection

We sourced real-world, first-person driving videos and 2D object tracks from the Waymo Open Perception Dataset [35], which contains diverse, urban driving scenarios recorded in San Francisco, Phoenix, and Mountain View, USA. We manually down-selected 142 scenarios, where an object (vehicle, bicycle, or pedestrian) is undergoing one of 6 potentially safety-critical traffic maneuvers (crossing-road, turning-in-road, lane-change, braking, overtaking, and entering/exiting a vehicle). Fig. 3(a) shows sample stimuli from all traffic maneuvers.

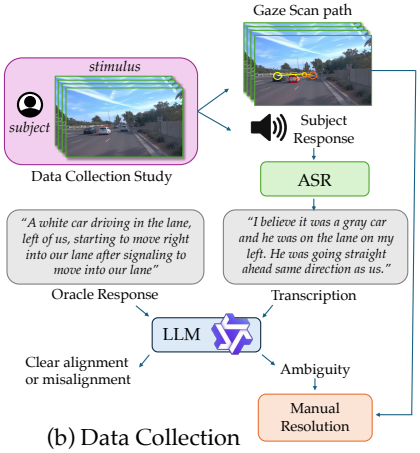
The final stimulus is a $3-5$ second clip from each selected scenario ending exactly when the maneuver is initiated. Furthermore, to study how long it takes for drivers to gain different levels of awareness, we extended each clip for a specified time offset *after* the maneuver begins, appending 0, 0.1, 0.3, 0.6, 1.0, 1.5, or 2 additional seconds. The final mean stimulus length was 4.7 seconds.

B. Collection Protocol

We collected awareness responses from subjects by querying them after presenting the visual stimulus. At the end of each stimulus, the screen was blanked and a purple mask appeared at the location of the target object (the object undergoing the traffic maneuver). Subjects were asked to verbally



(a) Stimuli Types



(b) Data Collection Pipeline

Fig. 3: **Data collection**. (a) We curated a set of video stimuli containing target objects (pedestrians and cyclists - green bounding box, and vehicles - red bounding box) that were performing traffic maneuvers in a variety of urban driving scenes. (b) Verbal responses to SAGAT probes were processed through a semi-automated pipeline, using automatic speech recognition and an LLM, to generate layered SA labels coupled to video stimuli and gaze scanpaths.

respond to: *"Was there a vehicle or pedestrian in the region marked by the purple highlight in the video? Do your best to describe it, what it looks like, what it's doing, and where it's headed. If you don't know or are guessing please say so."*

The study was conducted on a 27-inch Full-HD monitor, with verbal responses collected using a noise-canceling headset. Eye gaze was recorded at 60 hz during stimulus playback using a Tobii Pro Spark, mounted at the bottom of the screen. Each subject underwent an instructional video, eye gaze calibration, and four practice stimuli to familiarize them with the task. During the main study, each subject was shown at most 90 unique stimuli. Only one time offset was used per subject so that no stimulus was seen twice by a single subject. Additionally, each stimulus was shown at each particular time offset to at most four subjects.

The experimental procedure was approved by the WCG IRB (Protocol #20241945). Written informed consent was obtained from all subjects prior to participation. Our subject pool consisted of 21 self-identified male and 21 self-identified female subjects. Based on self-reported ages, 9 participants were below 30 years of age, 24 were 30-50, and 9 were over 50. All subjects possessed a valid US driver's license.

C. Data Collection Pipeline

An overview of our data pipeline is shown in Fig. 3(b). We first removed instances where audio transcriptions were illegible or not-recorded or for subjects where eye gaze was excessively noisy (with a large proportion of missing data). This yielded $3,320$ data points, each with a visual stimulus, a long-form text awareness-response (transcribed using Whisper [36]), and an eye gaze trace.

Next, we obtained a set of oracle responses for all 142 stimuli. The oracle responses are canonical responses capturing SA at all levels. The oracle watched the stimuli offset by the full two seconds, to provide maximum information about the target object, and were allowed to re-watch the stimuli if needed.

We compared the subject awareness-responses against the oracle responses to identify the binary layered SA labels from the response. Next, we attempted to use LLMs to extract the layered SA labels from responses by thorough comparison with the oracle response [37]. However, subject responses were sometimes found to be ambiguous, either due to a lack of descriptive detail, human error, or noisy transcriptions due to strong accents. For example, certain responses misidentified the color/type of the query vehicle, despite correctly identifying context and future awareness cues. This made direct LLM-based label extraction erroneous.

To achieve better ground truth accuracy, we therefore first used the LLM to filter ambiguous responses, by identifying responses that were fully aligned or misaligned with the oracle. We kept the LLM extracted labels for the clear cases, and manually corrected the awareness labels only from the ambiguous responses. The manual correctors had access to the gaze traces and the stimuli to ensure sound judgment. Out of $3,320$ total responses, 831 (25%) were found to be ambiguous.

D. Overall Dataset Statistics

Our final dataset consists of $3,320$ samples across 142 unique stimuli and awareness-responses, and eye gaze from 42 subjects. The average word count of the response transcriptions is 38.2 (vs. 33.8 for the oracle). In-terms of SA levels, 56.1% $(1,863)$ responses show awareness at all levels, 23.2% (769) of the responses show only existence and context awareness, 2.3% (75) of responses show only existence awareness, and 18.4% (611) responses are completely unaware. The low number of samples with existence-awareness only shows that perception and context awareness are closely linked.

We also analyze the temporal evolution of layered SA. In Fig. 4, we plot the average awareness of subject-responses against the time-offsets. According to the general intuition that more time should allow subjects to gain awareness [38], we observe that generally SA increases with time. Across

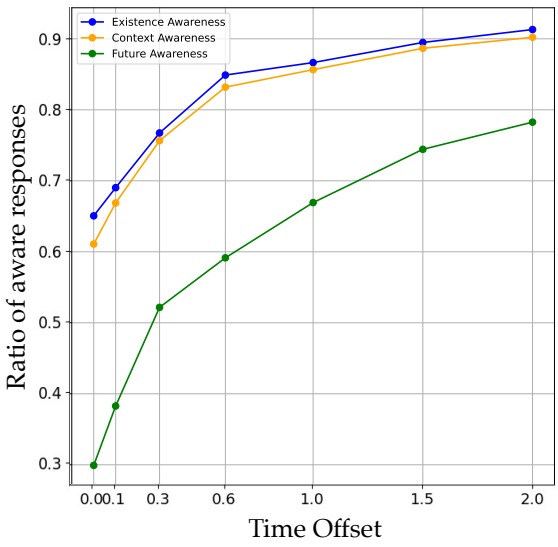


Fig. 4: SA of subjects sharply increases first, followed by a more gradual increment with larger time-offsets. Note that the progression of existence and context awareness with time is closely aligned.

all awareness levels, the curves show a sharp initial rise followed by a more gradual increase. This indicates that most subjects acquire awareness within the first few seconds, while the slower later growth likely reflects cases where subjects were initially distracted and required more time to register the target-object. Additionally, we observe that existence awareness is closely aligned with context awareness. Due to fewer existence-only samples and the alignment between existence and context awareness we combined existence and context awareness into 'present awareness'

IV. Layered Situational Awareness Estimation

For estimating a subject's layered SA of a target object given a driving scenario, or their eye gaze, the goal is to predict the subject's SA pertaining to the present (existence and context) and future projection of the target object.

A. Problem Formulation and Notation

We cast layered SA estimation as a multi-class classification problem. Given a visual stimulus $\mathcal{V}_t$, gaze scanpath $\mathcal{G}_t$, and a target object $\mathcal{O}_t$ from the stimulus, our goal is to estimate the awareness status, $\mathcal{A}_t$, where $\mathcal{A}_t \in$ {unaware, present-aware, future-aware}. Note that a sample is considered present-aware if it is existence or context-aware.

Visual Stimulus Representation: We represent the visual stimulus $\mathcal{V}_t$ as a RGB video,

$$\mathcal{V}_t = (\mathcal{I}_1, \mathcal{I}_2, \ldots, \mathcal{I}_t), \quad \mathcal{I}_k \in \mathbb{R}^{H \times W \times 3}, \tag{1}$$

where $\mathcal{I}_k$ denotes the kth frame in the video, and t denotes the number of frames in the video.

Eye Gaze Representation: We represent the eye gaze $\mathcal{G}_t$ as a temporal sequence of 2D coordinates over the visual stimulus.

$$\mathcal{G}_t = (g_1, g_2, \ldots, g_t), \quad g_k \in \mathbb{R}^{L \times 2}, \tag{2}$$

where g_k denotes the kth 2D gaze points at time k. L is the ratio of the frequency of the gaze-tracker (60Hz) and the video frame-rate (10Hz). Furthermore, we exclude saccade points from $\mathcal{G}_t$. Saccades are quick movements between fixations, and do not contribute to awareness [39]. We detect saccade movements by thresholding the velocity between gaze points.

Target Object Representation: We represent the target object $\mathcal{O}_t$ as a stream of 2D bounding box coordinates across the visual stimuli.

$$\mathcal{O}_t = (o_1, o_2, \ldots, o_t), \quad o_k \in \mathbb{R}^4, \tag{3}$$

where o_k denotes the bounding box coordinates (center x, center y, width, height) of the target object in $\mathcal{I}_k$.

B. Hand-Crafted Features for layered SA Estimation

Prior work [7], have studied existence-only SA with certain hand-crafted high-level features. In this section, we adapt existing features, and describe additional features that are relevant for layered SA estimation.

Spatial interaction between eye gaze points and the target object (G): Humans gain information through gaze fixations. Hence, the distance between the gaze fixation points and the target object should inform awareness. 'Gaze-distance' is defined as the minimum Euclidean distance between the 2D gaze fixations and the edges of the 2D bounding-box of the target object. As features we use (1) the gaze-distance at the end of the visual stimuli, (2) the average, and (3) the minimum gaze-distance throughout the visual stimuli.

Spatial and appearance properties of the target object (O): Next, we capture appearance and spatial properties of the target object. Specifically, we use (1) the duration for which the object was visible in the stimulus, (2) the contrast of the object against the background, (3) the target object's proximity to the ego vehicle, as approximated by the L2 distance between the object center and and (4) the size of the target object in pixels. Higher values for all these features may make it easier to become aware of the target object.

Target object presence in different retinal regions (R): Humans process visual information gained through each retinal region differently [7]. Thus, we model features based on when, and for how long, the target object persists in the foveal (R_1 = 2.5°), parafoveal (R_2 = 4.1°), perifoveal (R_3 = 9.1°), and macular (R_4 = 15.0°) regions. Specifically, we capture (1) time-elapsed since the target object was in each retinal region, (2) total time for which the target object was in each retinal region, and (3) the average gaze distance when the target object was within each retinal region. The target object is said to be in a given region when the gaze distance is within the radius (R_i) range of that region.

Influence of distractor objects (D): The above feature categories are adapted from prior work [7]. They do not explicitly encode how the presence of other objects in the scene influence the driver's awareness, thus we define new features.

- Number of distractor (non-target) objects in different retinal regions: We specifically study (1) average, (2) minimum and (3) maximum number of distractor objects

Method	Precision	Recall	F1-score	ROC-AUC
Random	0.32 ± 0.01	0.31 ± 0.01	0.29 ± 0.01	0.48 ± 0.01
Multiclass	0.56 ± 0.02	**0.60 ± 0.02**	0.56 ± 0.02	**0.76 ± 0.02**
Cascade	0.57 ± 0.02	0.58 ± 0.02	**0.57 ± 0.02**	0.76 ± 0.02
Hierarchical	**0.57 ± 0.02**	0.53 ± 0.02	0.54 ± 0.02	0.73 ± 0.01

TABLE I: Comparison of baseline approaches for layered SA estimation. Although the differences between present and future awareness are subtle and nuanced, the baseline results indicate that the high-level do contain some meaningful predictive cues for distinguishing between these awareness states.

Method	Precision	Recall	F1-score	ROC-AUC
G	0.46 ± 0.08	0.36 ± 0.01	0.33 ± 0.01	0.61 ± 0.01
R	0.46 ± 0.10	0.35 ± 0.01	0.32 ± 0.01	0.62 ± 0.01
O	0.51 ± 0.03	0.37 ± 0.02	0.36 ± 0.02	0.63 ± 0.01
D	0.28 ± 0.01	0.33 ± 0.01	0.29 ± 0.01	0.57 ± 0.02
all	0.57 ± 0.02	0.53 ± 0.02	0.54 ± 0.02	0.73 ± 0.01
all w/o G	0.55 ± 0.03	0.49 ± 0.02	0.50 ± 0.02	0.72 ± 0.02
all w/o R	0.56 ± 0.03	0.52 ± 0.02	0.53 ± 0.02	0.73 ± 0.02
all w/o O	0.51 ± 0.02	0.45 ± 0.02	0.46 ± 0.02	0.68 ± 0.01
all w/o D	0.54 ± 0.02	0.49 ± 0.02	0.51 ± 0.02	0.71 ± 0.02

TABLE II: We ablate the choice of feature groups used for predicting layered SA. The top-half shows that individually all features perform similarly, except distractor features (D), as they only capture non-target object features. The bottom-half shows that removing Object features (O) has the largest drop, and retinal region features (R) has negligible effect.

in each retinal region across the entire duration. We also use (4) the number of distractor objects in each retinal region at the pause.

- The number of distractor objects the gaze fixated on: We approximate this by (5) counting how many unique distractor objects had overlapping 2D gaze fixations within their bounding boxes.

These features coarsely approximate how the presence of distractor objects in the scene affects SA of a target object.

C. Baselines:

We train three simple baseline models, each based on the human-interpretable features described above, to assess performance at Layered SA estimation.

Multi-class Classifier: A single three-class support vector machine (SVM) classifier over the hand-crafted features.

Probabilistic Cascading of Binary Classifiers: The simple multi-class classifier ignores the hierarchical structure within different levels of SA. Thus, in this baseline, we cascade binary classifiers, one for each level of awareness, to get the final class probabilities. The final joint probability distribution is computed sequentially. Let P_i, $i \in P, F$ denote the individual classifier probabilities for present and future.

$$p_1 = P_{\text{P}}(Y = 1), \quad p_2 = P_{\text{F}}(Y = 1)$$

Then, the cascaded three-class probability distribution is:

$$P(Y = y) = \begin{cases} 1 - p_1, & \text{if } y = \text{unaware}, \\ p_1\,(1 - p_2), & \text{if } y = \text{present-aware}, \\ p_1\,p_2 & \text{if } y = \text{future-aware}, \end{cases}$$

Hierarchical Binary Classifiers: Here, the predictions from the preceding awareness level classifier is used as an additional input feature to the subsequent level classifier. Similar to the probabilistic cascaded classifiers, the final joint probability distribution is computed sequentially.

D. Experiments

We perform layered SA estimation using the 3 baseline approaches, on our dataset. We evaluate each baseline using repeated random sub-sampling. For each round, we create a test set through stratified random sampling with 20% of the samples from each class. We repeated this for 10 rounds and report mean and standard deviations across all metrics. We report F1-score, Precision, Recall and ROC-AUC. We report unweighted averages of per-class Precision, Recall and F1-score, to adjust for the class imbalance.

To what extent is layered SA predictable?

Tab. I summarizes baseline performances for layered SA estimation. Distinguishing present from future awareness using only gaze scanpaths and scene context is inherently subtle and requires nuanced reasoning (Fig. 2). Nevertheless, even with simple handcrafted features, we can achieve substantial gains over random performance, improving Precision by 0.25, Recall by 0.29, F1-score by 0.28, and ROC-AUC by about 0.28.

Despite these improvements, there remains significant headroom. Gaze-fixations do not directly cause awareness, in many instances humans can gaze over relevant objects without gaining awareness [40]. Moreover, how humans derive awareness from peripheral vision is complex and not fully captured by coarse features such as the duration and position of objects across retinal regions [41].

Which features contribute most towards layered SA estimation?

We perform ablation experiments with different combinations of the Gaze-Object (G), Object-Properties (O), Retinal Region (R), and Distractor (D) features, as described in Sec. IV-B. We report the results in Tab. II. The top-half of the table, evaluates layered SA estimation using individual feature groups, to study their individual relevance. The performance across individual feature groups is comparable, however it is worse compared to the combination. Individual feature groups show atleast 0.1 points worse on precision, 0.2 on recall and F1-score, and 0.16 worse on ROC-AUC. Note that the performance with Distractor features only is considerably worse because they do not capture any information about the target object.

The bottom-half of Tab. II, evaluates the performance with each feature group removed. Performance shift very slightly when Gaze-Object (G) features are removed, precision drops by 0.02, recall by 0.04, F1-score by 0.04 and the ROC-AUC Score by 0.01 points. Removing Distractor (D) features results in a similar small drop in performance. Performance remains comparable when the Retinal Region (R) based features are removed. Removing Object-Property (O) features degrades performance the most. Precision drops by 0.06, recall by 0.08, F1-Score by 0.08 and ROC-AUC by 0.05. These results

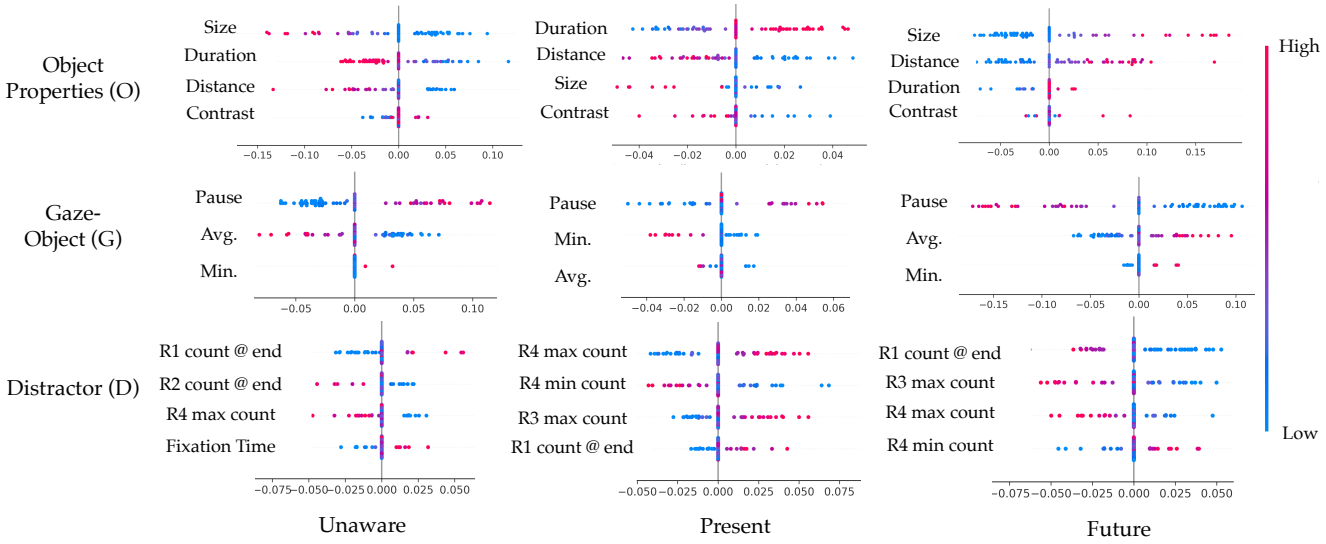


Fig. 5: SHAP values for O, G, D. Red/blue denote high and low feature values. Points on the left contribute negatively towards class prediction and vice-versa. Lower object size contributes towards unawareness and bigger sizes towards future-awareness. Similarly, larger distances between objects and gaze point at the end, contribute positively to future awareness and negatively to unawareness. Finally, more distractor objects impair future-awareness.

suggest that Object-Properties, Gaze-Object interaction, and Distractor features contribute towards overall performance.

We also perform a SHAP analysis [42] of the Multiclass SVM to understand how features within each feature group contribute towards layered SA estimation. We pick a class-wise balanced subset of the training and test set. Fig 5 shows the per-class shap values for the Object Property (O), Gaze-Object (G), and Distractor (D) feature groups.

We study Object-Properties (O) in the first row of Fig. 5. Bigger object sizes negatively influences unaware and present awareness but positively influences future awareness. Longer durations cause present and future-awareness. Interestingly, smaller distance to the ego vehicle cause unawareness, and larger distances increase future awareness. Contrast has the lowest influence, and behaves un-intuitively.

Next, we look at the Gaze-Object (G) features. The distance between the object and the gaze fixation at the end contributes the most. Higher distance at the end contributes to unawareness, the trends for present awareness are similar but weaker, and for future awareness the trend is flipped, where-in the lower distance contributes to future awareness.

Finally, we look at the Distractor (D) features. We plot 4 features that contribute the most out of the 17 distractor features (4 per retinal region + 1 (total non-target fixations)). More distractor objects in the smallest retinal region (R_1) at the end contribute more towards unawareness. Fewer distractor objects in non-foveal regions contribute positively towards unawareness. Interestingly, more distractor objects in non-foveal regions contribute positively towards present-awareness. But the trends are flipped in future-awareness. This shows that distractor objects are more likely to impair future-awareness.

V. LIMITATIONS

We highlight a few limitations of our work. Firstly, while our dataset exposes only gaze information from drivers, gaze alone is insufficient to fully determine awareness. Humans routinely miss important visual information that is "right in front of our eyes" [43], often rely on peripheral vision [41]. Personalization to individuals, and multi-modal inputs (including, for example, driving control input) are likely to reduce this gap. Secondly, while our dataset serves as a useful means to study SA at different levels from gaze scanpaths, it may lack the ecological validity of other approaches such as DART [44], which collect data during real world driving. This makes the real-world applicability of our approach challenging. However, our approach has the advantage of supporting the study of multiple subjects on highly diverse stimuli, with multiple repetitions, such as shown in Fig. 2, and as such can provide useful insights into the mechanisms of SA that prior works do not. Finally, we cast SA estimation as an information impoverished multi-class classification problem, as opposed to directly reasoning in language. The modest size of our dataset, makes Vision/Language model finetuning challenging

VI. CONCLUSIONS

We have presented a new dataset for studying the situational awareness of drivers using rich natural language feedback to assess SA at multiple levels. Using our dataset, we analyzed how a set of intuitive high-level eye gaze and scene-context features can inform SA, and additionally provided simple baselines to predict driver SA. By making our data and baselines available for further use, we hope to provide a community benchmark for an important problem in intelligent vehicle research.

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